

An inequality between determinant and signature for L -space knots

Machine-generated conjecture, its checks, a partial proof, and the open case

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Abstract

We consider the inequality $\det(K) \leq 1 + |\sigma(K)|$ for L -space knots. It was produced by a program that fits linear inequalities to the KnotInfo database and discards what its automatic checks refute. It survived those checks and the computations of Section 3. We prove it for every L -space knot that is an iterated torus knot, in particular for all algebraic knots. The hyperbolic case is open, and it is where the inequality is sharp: equality holds on every member tested of an infinite family of hyperbolic, non-braid-positive L -space knots. In a second hyperbolic family, added after four rounds of review, equality never occurs and the slack is bounded below, so sharpness is a property of the first family rather than of the hyperbolic case in general. Sections 2 and 3 give the provenance and the computations, so that the proof in Sections 5 and 6 can be checked without our files; the one lemma we prove from scratch rather than cite is additionally formalised in Lean 4 with no `sorry`.

1 Statement and conventions

Signs follow KnotInfo: the right-handed trefoil has $\sigma = -2$, $s = 2$, $\tau = 1$. A knot $K \subset S^3$ is an *L -space knot* if some positive Dehn surgery on K yields an L -space. Positive torus knots then have $\sigma < 0$. We do not assume a sign for a general L -space knot: $\sigma \leq 0$ is carried through the induction of Theorem 1 for the knots treated there, and it held on every L -space knot we computed (all 632 census knots and all 68 knots of the families of Section 3). Write $\det(K) = |\Delta_K(-1)|$ and $g = g(K)$ for the Seifert genus.

Conjecture 1 (K33). *For every L -space knot K ,*

$$\det(K) \leq 1 + |\sigma(K)|.$$

We use the following standard facts. An L -space knot is fibered, strongly quasipositive, and satisfies $g = g_4 = \tau$ (Ozsváth–Szabó, Ni, Hedden); its Alexander polynomial has the form $\Delta_K(t) = \sum_{k=0}^{2n} (-1)^k t^{a_k}$ with $a_0 = g > a_1 > \dots > a_{2n} = -g$ (Ozsváth–Szabó). Consequently $|\sigma(K)| \leq 2g$ and $\det(K) \leq 2g + 1$.

2 Provenance

The inequality was not conjectured by a person. It is candidate K33 of 42 statements emitted by `ragusa`, an engine that enumerates bounds of the form (property) $\Rightarrow$ (target $\leq c \cdot$ feature $+k$) over a table of invariants, here KnotInfo restricted to prime knots with 3 to 13 crossings, and reports what survives its filters together with the evidence behind each survivor. K33 was found on the `1_space` column, where the table contains 11 knots; those knots were the whole of its initial evidence.

The other 41 candidates were triaged in the same session: 18 were refuted by explicit knots, 13 turned out to be known theorems or true by definition, 3 had one clause refuted and one left open, and 6 remain open but rest on weak evidence. Two remained open with evidence beyond the table: K33, and $u = g$ for L -space knots.

3 Verification history

All computations use SnapPy (knot Floer homology, for the L -space property and τ), Sage (Seifert matrices, signatures, determinants) and exact formula implementations; scripts are listed in Section 9.

family of L -space knots	tested	viol.	how the invariants are known
iterated torus knots	11728	0	exact: cabled Alexander polynomial, Litherland's formula
SnapPy census L -space knots (all)	632	0	Seifert matrix of a positive braid word
census knots, independent route	271	0	knot Floer homology, Sage signature
non-braid-positive families	68	0	Seifert matrix of the braid word
one-bridge braids $B(w, b, t)$, $w \leq 15$	274	0	Seifert matrix of the braid word
twisted torus knots $K(p, q; r, s)$	549	0	knot Floer homology, then exact Seifert signature

The census braid words are those published by Baker and Kegel (arXiv:2203.12013, Appendix), which cover all 632 census L -space knots. Every closure was checked to be the named census knot: 631 of 632 by SnapPy's isometry test directly, the remaining one after retriangulation and at higher precision. All 632 signatures are exact, computed by congruence diagonalisation over $\mathbb{Q}$ from the integral matrix $V + V^T$; they agree with an earlier floating-point computation on all 632 knots and with Sage's own exact signature on the 511 knots where that finished. The one-bridge braids are L -space knots by [10]; for the 40 smallest this was also confirmed directly by knot Floer homology, with the genus matching the braid-surface genus.

Twisted torus knots. The family $K(p, q; r, s)$, the closure of $(\sigma_1 \cdots \sigma_{p-1})^q (\sigma_1 \cdots \sigma_{r-1})^s$, was added after four rounds of review and meets none of the families above. Membership of the L -space class was decided for each knot by its own knot Floer homology rather than by a classification, which leaves 549 L -space knots of genus up to 18, of which 171 are hyperbolic and 215 have a non-positive braid word and so fall outside the positive-braid case. There are 0 violations. On this family the determinant is computed twice over, once from the Alexander polynomial read off the knot Floer gradings and once as $|\det(V + V^T)|$ from the Seifert matrix; the two agree on all 549 knots.

Independent check. A language model with no access to our computations re-derived the lattice count and recomputed, independently: the six torus signatures used here, Lemma 1 and the bound on $N_<$ for all coprime $p \leq 24$, $q < 120$, the closed forms and their reductions, the list of leftover small pairs in the induction, the induction step under worst-case hypotheses, and the identity (1) on three examples. It found no errors. The gaps it flagged (an unsupported sign assertion, boundary points in the count, the $p < q$ convention, the nontriviality hypothesis in (F4), the derivation of the symmetry of S , missing references) are addressed in this version. A third model corrected the proof of Lemma 1, where an earlier version claimed an equivalence that is only a sufficient condition, and asked for the hypotheses and chirality conventions now stated; the identity $\sigma = 4N_< - 2g$ was verified

on all coprime $p \leq 15$, $q < 50$, where $N_{<}/g \leq 0.234$, so the branch proved here is the operative one. A second model, asked to attack the open case, contributed the congruence of Section 5, the one-bridge braid family of the table, and the rigidity observation of Section 7; its computations of the one-bridge braids (274 knots, 0 violations, 70 equalities) and of the Baker–Kegel invariants were reproduced independently here.

The hypothesis is not redundant. Evaluating the same inequality on census knots whose braid word is positive or negative but which are *not* L -space knots produces 25 violations. So the L -space condition does the work; this is unlike several of the 42 candidates, whose hypotheses turned out to be irrelevant.

Sharpness. Equality $\det = 1 + |\sigma|$ holds for $T(2, n)$, for the $(2, q)$ -cables of the trefoil, and, among the families tested beyond the census, for 62 of 68 knots, including every Baker–Kegel K_n with $n = 1, \dots, 25$. These are hyperbolic L -space knots that are not braid positive. The twisted torus family behaves differently: 127 of its 549 L -space knots attain equality and not one of them is hyperbolic, and over its 171 hyperbolic members the slack $1 + |\sigma| - \det$ is at least 4.

4 A reformulation

For an L -space knot let $S \subset \mathbb{Z}_{\geq 0}$ be its staircase set (the *formal semigroup*; for an algebraic knot it is the semigroup of the singularity, but for a general L -space knot it need not be closed under addition), the complement of the g gaps, determined by $t^g \Delta_K(t) = (1 - t) \sum_{s \in S, s < 2g} t^s + t^{2g}$. Evaluating at $t = -1$ and using the symmetry $s \leftrightarrow 2g - 1 - s$ between $S \cap [0, 2g)$ and the gaps, which follows from $\Delta_K(t) = \Delta_K(t^{-1})$ and gives $|S \cap [0, 2g)| = g$ as well, one obtains

$$\det(K) = |2D + 1|, \quad D = \#\{\text{odd gaps}\} - \#\{\text{even gaps}\}. \quad (1)$$

Hence Conjecture 1 is equivalent to $|\sigma| \geq 2D$ when $D \geq 0$, and to $|\sigma| \geq 2|D| - 2$ when $D < 0$. Identity (1) was checked on 11728 iterated torus knots with 0 mismatches.

Remark 1. The cruder route fails. Since the coefficients of Δ_K are ± 1 , $\det(K)$ is at most the number of nonzero coefficients, so Conjecture 1 would follow from $|\sigma| \geq \#\{\text{terms}\} - 1$. That inequality is false: it fails on 401 of the 11728 iterated torus knots tested, and the failures are exactly the knots where Conjecture 1 is sharp. Any proof must therefore see the cancellation in $\Delta_K(-1)$.

5 An arithmetic refinement

For any knot, $\Delta_K(-1) \equiv 1 \pmod{4}$ and $\text{sign } \Delta_K(-1) = (-1)^{\sigma/2}$ (Murasugi), so $\det(K) \equiv (-1)^{\sigma/2} \pmod{4}$ and therefore

$$1 + |\sigma(K)| - \det(K) \equiv 0 \pmod{4}. \quad (2)$$

Checked on all 1523 L -space knots computed here with an exact signature and determinant: 0 failures. Two consequences. First, a counterexample to Conjecture 1 must satisfy $\det(K) \geq |\sigma(K)| + 5$; the inequality cannot fail narrowly. Second, combining (2) with (1), the *signature defect* $2g - |\sigma|$ of an L -space knot satisfies $2g - |\sigma| \equiv 0 \pmod{4}$ when $D \geq 0$, and Conjecture 1 becomes

$$2g - |\sigma| \leq 4\#\{\text{even gaps}\} \quad (D \geq 0),$$

both sides being multiples of 4. (For $D < 0$ the corresponding statement is $2g - |\sigma| \leq 4 \# \{\text{odd gaps}\} + 2$.) In particular the conjecture holds automatically for every L -space knot with $\det \leq 3$ and for every definite one.

6 Proof for iterated torus knots

Throughout, $p \geq 2$ and q are coprime, $T(p, q)$ is the positive torus knot with $g(T(p, q)) = (p-1)(q-1)/2$, and $K_{p,q}$ denotes the (p, q) -cable of K (winding number p). We use:

- (F1) $\Delta_{K_{p,q}}(t) = \Delta_K(t^p) \Delta_{T(p,q)}(t)$; since $\Delta_K(1) = 1$ this gives $\det(K_{p,q}) = \det(K) \det(T(p, q))$ for p odd and $\det(K_{p,q}) = \det(T(p, q))$ for p even.
- (F2) Litherland's cabling formula $\sigma(K_{p,q}) = \sigma_{(-1)^p}(K) + \sigma(T(p, q))$, so $\sigma(K_{p,q}) = \sigma(K) + \sigma(T(p, q))$ for p odd and $= \sigma(T(p, q))$ for p even.
- (F3) $\det(T(p, q)) = 1$ if p, q are both odd, $= p$ if q is even, $= q$ if p is even.
- (F4) (Hedden; Hom) for K nontrivial, $K_{p,q}$ is an L -space knot if and only if K is one and $q \geq p(2g(K) - 1)$.

(F1)–(F3) were verified on 4088 cable instances, with 0, 0 and 0 mismatches respectively.

Lemma 1. *For integers $2 \leq p < q$ with $\gcd(p, q) = 1$, $\sigma(T(p, q)) \leq -g(T(p, q))$, where $g(T(p, q)) = \frac{(p-1)(q-1)}{2}$; in particular $|\sigma| \geq g$.*

Proof. We use the lattice count for torus knot signatures (Brieskorn; Gordon–Litherland–Murasugi [6]; Litherland [14]), in the normalisation $\sigma(T(2, 3)) = -2$ of Section 1: $\sigma(T(p, q)) = N_{\text{out}} - N_{\text{in}}$ over the grid $G = \{1, \dots, p-1\} \times \{1, \dots, q-1\}$, where (i, j) is counted in N_{in} when $\frac{1}{2} < \frac{i}{p} + \frac{j}{q} < \frac{3}{2}$. The involution $(i, j) \mapsto (p-i, q-j)$ of G exchanges the two outside regions, so $N_{\text{out}} = 2N_{<}$ with $N_{<} = \#\{(i, j) \in G : \frac{i}{p} + \frac{j}{q} < \frac{1}{2}\}$. No lattice point of G lies on a boundary: $\frac{i}{p} + \frac{j}{q} \in \{\frac{1}{2}, \frac{3}{2}\}$ gives $2(iq + jp) \in \{pq, 3pq\}$, so $p \mid 2i$ and $q \mid 2j$, forcing $2i = p$ and $2j = q$ with p and q both even, contradicting $\gcd(p, q) = 1$. As $N_{\text{in}} + N_{\text{out}} = 2g$, the count reads $\sigma(T(p, q)) = 4N_{<} - 2g$, so $|\sigma| \geq g$ holds if $N_{<} \leq g/4$, and also if $N_{<} \geq 3g/4$. We prove the first branch, $N_{<} \leq (p-1)(q-1)/8$, which gives $\sigma(T(p, q)) \leq -g < 0$.

For fixed i the number of admissible j is $\lceil f(i) \rceil - 1 \leq f(i)$ with $f(i) = \frac{q}{2} - \frac{iq}{p}$, and $f(i) > 0$ exactly for $i < p/2$. Summing over those i : if p is even, with $m = p/2 - 1$, one gets $N_{<} \leq q(p-2)/8$, and $q(p-2) \leq (p-1)(q-1)$ reduces to $q \geq p-1$; if p is odd, with $m = (p-1)/2$, one gets $N_{<} \leq q(p-1)^2/(8p)$, and $q(p-1)/p \leq q-1$ reduces to $p \leq q$. Both hold because $q > p$. $\square$

The lattice inequality is machine-checked. The combinatorial core of this proof, the bound $N_{<} \leq (p-1)(q-1)/8$ for all integers $2 \leq p < q$, has been formalised in Lean 4 against Mathlib and carries no `sorry`: `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. Coprimality is not needed for it. The formalisation covers the column decomposition of the grid count, the per-column bound, the cut at $(p-1)/2$ in both parities, the Gauss sum, the closed forms, and the two final comparisons; it does not cover the topology the lemma is used with, namely the lattice count for $\sigma(T(p, q))$, which is cited. So the part of Lemma 1 proved here from scratch, rather than cited, is the part that is now checked by a machine. The file is `K33Lattice/Basic.lean`.

Every step of this proof was also checked numerically on 429 pairs with $p \leq 15$, $q < 60$ (0 failures), and the conclusion holds on all 2166 pairs with $p \leq 20$, $q < 200$. Lemma 1 says $-\sigma \geq b_1/2$ for torus knots, which is stronger than the linear bounds known for arbitrary positive braid links (Feller proved

a linear bound; Greene–Liechti give $-\sigma \geq b_1/4$, see the abstract of [5]). We did not find Lemma 1 in the literature; it may be folklore.

Theorem 1. *Every L -space knot which is an iterated torus knot satisfies $\det(K) \leq 1 + |\sigma(K)|$. In particular this holds for all positive torus knots, hence for their mirrors since $\det$ and $|\sigma|$ are mirror invariant, and for all algebraic knots, which are iterated torus knots with positive parameters [3] and hence L -space knots.*

Proof. The induction is on the cabling depth, with torus knots at depth one; by (F4) applied to the companion (Hom’s “only if”), every companion of an L -space iterated torus knot is again an L -space knot, so the hypothesis is available at each step. We prove simultaneously that $\det(K) \leq 1 + |\sigma(K)|$ and $\sigma(K) < 0$. For torus knots we may assume $p < q$, since $T(p, q) = T(q, p)$; in a cable $K_{p,q}$ of a nontrivial L -space knot, (F4) gives $q \geq p(2g_K - 1) \geq p$, and $\gcd(p, q) = 1$ excludes $q = p$.

Base case. For $T(p, q)$, $\sigma < 0$ is part of Lemma 1, and: if p, q are both odd then $\det = 1$. If q is even then $\det = p$ and Lemma 1 gives $|\sigma| \geq 3(p - 1)/2$ (using $q \geq 4$), whence $1 + |\sigma| \geq p$. If p is even then $\det = q$; for $p = 2$ we have $|\sigma(T(2, q))| = q - 1$, so $1 + |\sigma| = q$, with equality; for $p \geq 4$, Lemma 1 gives $|\sigma| \geq 3(q - 1)/2 \geq q - 1$.

Induction step. Let K be an L -space knot with $\det(K) \leq 1 + A$, where $A = |\sigma(K)| \leq 2g_K$, and let $K_{p,q}$ be an L -space cable, so $q \geq p(2g_K - 1)$ by (F4). Put $B = |\sigma(T(p, q))|$. By the induction hypothesis $\sigma(K) < 0$, and $\sigma(T(p, q)) < 0$ by Lemma 1, so (F2) gives $\sigma(K_{p,q}) < 0$ and $|\sigma(K_{p,q})| = A + B$ for p odd, $= B$ for p even; this also propagates the sign claim.

If p is even, then $\det(K_{p,q}) = \det(T(p, q)) = q$ by (F1) and (F3), and $q \leq 1 + B$ was shown in the base case.

If p and q are both odd, then $\det(K_{p,q}) = \det(K) \leq 1 + A \leq 1 + A + B$.

If p is odd and q even, then $\det(K_{p,q}) = p \det(K) \leq p(1 + A)$, and the claim $p(1 + A) \leq 1 + A + B$ is $(p - 1)(1 + A) \leq B$. By Lemma 1 it suffices that $1 + A \leq (q - 1)/2$, i.e. $q \geq 2A + 3$, and since $A \leq 2g_K$ it suffices that $q \geq 4g_K + 3$. As $p \geq 3$, (F4) gives $q \geq 6g_K - 3$, which is $\geq 4g_K + 3$ exactly when $g_K \geq 3$. For $g_K \in \{1, 2\}$ the uncovered pairs are finite: for $g_K = 1$ they are $(p, q) = (3, 4)$ and $(5, 6)$, where $B = 6 \geq 6$ and $B = 16 \geq 12$; for $g_K = 2$ it is $(3, 10)$, where $B = 14 \geq 10$. $\square$

The induction step was also verified directly, with the worst admissible values $A = 2g_K$ and $\det(K) = 1 + A$, over $g_K \leq 11$, $p \leq 11$ and the full range of q allowed by (F4): 0 failures.

7 The open case

Theorem 1 does not reach the hyperbolic L -space knots, and that is where the inequality is sharp. Baker and Kegel (arXiv:2203.12013) construct hyperbolic L -space knots K_n , the closures of the braid words

$$[(2, 1, 3, 2)^{2n+1}, -1, 2, 1, 1, 2],$$

which are not braid positive for $n = 1$ and are not known to be braid positive for $n \geq 2$; Himeno (arXiv:2506.22934) constructs infinitely many hyperbolic L -space knots that are provably not braid positive. Equality $\det = 1 + |\sigma|$ holds for every Baker–Kegel K_n we tested, $n = 1, \dots, 25$.

Question 1. *Does $\det(K) \leq 1 + |\sigma(K)|$ hold for every L -space knot, in particular for the hyperbolic ones? Equivalently, by (1), is $|\sigma(K)| \geq 2(\#\{\text{odd gaps}\} - \#\{\text{even gaps}\})$?*

The sharp case carries a rigidity statement. If all g gaps are odd then $S \cap [0, 2g)$ consists of the even numbers, so $\Delta_K = \Delta_{T(2, 2g+1)}$, $D = g$ and $\det = 2g + 1$; Conjecture 1 then forces $|\sigma(K)| = 2g$, that is, the symmetrised Seifert form is definite. So the conjecture implies: every L -space knot with the Alexander polynomial of $T(2, 2g + 1)$ has $\sigma = -2g$. This is a weak form of Problem 1.21(c)(i) of the K3 problem list [11] (knot Floer homology detects $T(2g + 1, 2)$), known there for $g \leq 2$ by Ghiggini and by Farber–Reinoso–Wang. Any proof of Conjecture 1 therefore proves this definiteness statement.

What is known about the sharp case. Three further facts, contributed by a fourth reader and checked here against the exact data. (i) Such a K is not a nontrivial cable: the staircase set of a cable $J_{p,q}$ is $\{ps + qk : s \in S_J, 0 \leq k \leq p - 1\}$, whose smallest positive element is $\min(p s_{\min}, q) \geq 4$, whereas $S_{T(2, 2g+1)}$ has smallest positive element 2; here $s_{\min} \geq 2$ because 1 is a gap of every nontrivial L -space knot, which holds on all 11728 iterated torus knots computed. Among those knots, the only ones with $\Delta = \Delta_{T(2, 2g+1)}$ are the $T(2, 2g + 1)$ themselves. (ii) The roots of $\Delta_{T(2, 2g+1)}$ are simple and on the unit circle, so the Levine–Tristram function has g jumps of ± 2 on the upper half circle and $\sigma(K) = -2g + 4m$, where m counts the positive jumps; the question is exactly $m = 0$, that is, that the signature function of K agrees with that of $T(2, 2g + 1)$. (iii) Definiteness does not by itself force the torus knot: among the 11889 L -space iterated torus knots computed, exactly 21 are definite, namely the $T(2, n)$ together with $T(3, 4)$ and $T(3, 5)$ (the knots among the ADE singularity links). Neither of the last two has the Alexander polynomial of a $T(2, 2g + 1)$, so on this restricted class definiteness and detection still coincide.

The hypothesis of the sharp case is never met by anything else. Across every L -space knot for which a determinant, a genus and an exact signature were computed here — 1523 of them, from the census, the one-bridge braids, the families beyond the census and the twisted torus knots, reaching genus 145 — exactly 125 satisfy $\det = 2g + 1$, equivalently $\Delta = \Delta_{T(2, 2g+1)}$. Every one of them has $\sigma = -2g$, as Conjecture 1 requires, and every one of them *is* a $T(2, 2g + 1)$: each was put into braid form and found to be the closure of σ_1^{2g+1} in B_2 , which identifies the knot outright. The largest genus meeting the hypothesis is 7, against 145 in the sample. So on everything computed, the sharp case is not merely consistent with the conjecture, its hypothesis is empty outside the torus knots — which is the detection phenomenon of Problem 1.21(c)(i) appearing directly in the data.

Together these narrow a counterexample to Conjecture 1 in the sharp case: it would have to be hyperbolic or a satellite with a non-cable pattern, it could not be braid positive, and it would need a positive Levine–Tristram jump at one of the g roots.

Why the induction stops. For a torus knot the gaps of $\langle p, q \rangle$ satisfy $2g - |\sigma| = 4 \#\{\text{gaps} > pq/2\}$, so Theorem 1 for torus knots says $\#\{\text{gaps} > pq/2\} \leq \#\{\text{even gaps}\}$. No analogue of the half-conductor $pq/2$ is available for a hyperbolic L -space knot, and that is exactly what the cabling induction replaces.

Sharpness is not typical of the hyperbolic region. The Baker–Kegel family attains equality and is hyperbolic and not braid positive, which is what makes it the natural place to look for a counterexample. The twisted torus family of Section 3 is hyperbolic and not braid positive too, and there equality never occurs: every hyperbolic member has slack at least 4, while the members that do attain equality are exactly the non-hyperbolic ones. So equality is not a feature of the hyperbolic L -space knots in general, and the question is what distinguishes the Baker–Kegel family, not what distinguishes the hyperbolic case.

Two further questions. Which L -space knots attain equality? The known ones are $T(2, n)$, the $(2, q)$ -cables, the Baker–Kegel family, and 70 of the 274 one-bridge braids computed here. Does the inequality follow from a property of the symmetrised Seifert form of an L -space knot, its determinant being bounded by its signature? Definite strongly quasipositive links are studied in [2].

8 A second statement from the same run

The program produced one other statement that its checks could not refute and that we could not place in the literature: for every L -space knot, $u(K) = g(K)$, where u is the unknotting number.

Half of it is immediate. For an L -space knot $g = g_4$, and $u \geq g_4$ always, so $u \geq g$. The content is $u \leq g$, which holds for braid positive knots by Rudolph, hence for every L -space knot known to be braid positive; by Baker and Kegel [1] that is all but one of the census L -space knots. What is open is the non-braid-positive case, and there the only evidence is explicit unknottings.

Each certificate is a sequence of g crossing changes ending at the unknot, found by a search that keeps only the changes lowering $|\tau|$ by exactly one, which is necessary for an unknotting of that length. Certificates were produced for the census exceptions, for the Baker–Kegel and Himeno families up to genus 42, and, in this version, for the twisted torus knots: all 186 of those that are L -space knots with a non-positive braid word and genus at most 10 were certified, 31 of them hyperbolic, with no failure and no counterexample.

Unlike the inequality, this statement has not been shown to need its hypothesis. The half we can prove, $u \geq g$, uses the L -space property only through $g = g_4$, which already holds for every fibered strongly quasipositive knot; and we know of no fibered strongly quasipositive knot with $u > g$. Whether $u = g$ is really a statement about L -space knots, or about a wider class, is part of the question.

9 Reproducibility

Scripts, each writing the JSON or JSONL file the corresponding numbers are read from: `iterated.py` (exact iterated torus knots), `lemma_L.py` (all steps of Lemma 1), `induction_proof.py` ((F1)–(F3) and the induction step), `proof_steps.py` (identity (1)), `census_lspace_fast.sage` (all census L -space knots from the Baker–Kegel braid words), `census_sage_native.sage` (the non- L -space comparison), `families_fast.sage` (families beyond the census), `exact_signature.py` and `exact_census.py` (exact inertia for all census knots), `braid_identity.py` (the closures are the named census knots), `onebridge.sage` and `onebridge_lspace.py` (the one-bridge braid family), `twisted_torus.py` and `twisted_torus_sigma.sage` (the twisted torus family and its exact signatures), and `twisted_torus_ug.py` (the unknotting certificates of Section 8), `definiteness.py` and `definiteness_identify.py` (the sharp case across all families). Tools: SnapPy 3.3.2 with `knot_floer_homology`, Sage 10.7, khoca 1.5. The Lean formalisation of the lattice inequality is `K33Lattice/Basic.lean`, built against Mathlib at revision `0df444a360eaa60ab8c11dca51a86af692955474` with `leanprover/lean4:v4.33.1`.

Data provenance. The candidate was generated from KnotInfo (<https://knotinfo.org>); no KnotInfo data is reproduced here. The braid words for census knots are from Baker and Kegel’s appendix.

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